THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability Theory

Tutorial 3
2026/09/22

1. Suppose that an airplane engine will fail, when in flight, with probability $1-p$ independently from engine to engine. Assume that the airplane will make a successful flight if at least 50 percent of its engines remain operative. For what values of p are a four-engine plane preferable to a two-engine plane?

2. A coin having probability p of coming up heads is successively flipped until the r-th head appears. Suppose that the results of flips are mutually independent. Argue that X, the number of flips required, will be n with probability

$$\mathbb{P}(X=n)=\binom{n-1}{r-1}p^r(1-p)^{n-r},\quad n\geq r.$$

This is known as the *negative binomial distribution* with parameters p and r.

3. Suppose that independent trials, each of which results in any of m possible outcomes with respective probabilities $p_1,\ldots,p_m$, $\sum_{i=1}^m p_i=1$, are continually performed. Let X denote the number of trials needed until each outcome has occurred at least once.

   (a) If $m=2$, show that for any $n\geq 1$, we have

   $$\mathbb{P}(X>n)=(1-p_1)^n+(1-p_2)^n.$$

   (b) Find $\mathbb{P}(X>n)$ for general $m\geq 2$.

   (c) Find $\mathbb{P}(X=n)$.

   Hint: You may use the result in Homework 1 Question 1.

4. (a) Prove the memoryless property of geometric random variables. i.e., if $X\sim Geom(p)$, then

   $$P(X=n+k\mid X>k)=P(X=n),\quad \forall n,k\geq 0.$$

   (b) Prove the memoryless property of exponential random variables. i.e., if $X\sim Exp(\lambda)$, then

   $$P(X>n+k\mid X>k)=P(X>n),\quad \forall n,k\geq 0.$$